CO1

B. Venkata Jayani
192371005

CSA1305 - TOC

Task1

$\Sigma = \{0, 0\}$

Finite Strings: A sequence of symbols, eg: 1010

Empty string (ε): A string containing no symbols

Language: A set of strings over an alphabet, e.g.,

$L = \{0, 1, 11, 00\}$

$L = \{ w \in \{0,1\}^* : |w| \geq 1, \text{first}(w) = \text{last}(w) \}$:

Finite Automaton

$M = (Q, \Sigma, \delta, q_0, F)$.

Q: Set of states

Σ: Input Alphabet

δ: Transition function

q_0: Initial state

F: Set of final states

Role of components

States: Represent the current processing condition

Transitions: Define movement based on input.

Initial state: Starting point.

Final states: Indicate valid access codes.

Sample strings

0	Accepted
1	"
101	Accepted
1001	Accepted
0110	Accepted
01	Rejected
110	Rejected.

Task: 2

States: q_0 - initial

q_1 - {0, 0}

q_2 - {0, 1}

q_3 - {1, 1}

q_4 - {1, 0}

Five-tuple: $M = \{Q, \Sigma, \delta, q_0, F\}$

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

$\Sigma = \{0, 1\}$

$F = \{q_1, q_3\}$

$q_0 = q_0$

State Transitions: $q_0 \xrightarrow{0} q_1$

$q_0 \xrightarrow{1} q_3$

$q_1 \xrightarrow{0} q_1$; $q_1 \xrightarrow{1} q_2$

$q_2 \xrightarrow{0} q_1$; $q_2 \xrightarrow{1} q_2$

$q_3 \xrightarrow{0} q_4$; $q_3 \xrightarrow{1} q_3$

$q_4 \xrightarrow{0} q_4$; $q_4 \xrightarrow{1} q_3$

Transition table

State	0	1
$\rightarrow q_0$	q_1	q_3
$* q_1$	q_1	q_2
q_2	q_1	q_2
$* q_3$	q_4	q_3
q_4	q_4	q_3

Input Verification:

Input	State sequence	Result
101	$q_0 \rightarrow q_3 \rightarrow q_4 \rightarrow q_3$	Accepted
100	$q_0 \rightarrow q_3 \rightarrow q_4 \rightarrow q_4$	Rejected
0110	$q_0 \rightarrow q_1 \rightarrow q_2 \rightarrow q_1 \rightarrow q_2$	Rejected
1101	$q_0 \rightarrow q_3 \rightarrow q_3 \rightarrow q_4 \rightarrow q_3$	Accepted

Task 3

States : q_0 = initial state

q_1 = recognized 1

q_2 = recognized 10

q_3 = recognized 101 (final).

Five tuple : $M = (Q, \Sigma, \delta, q_0, F)$

$Q = \{q_0, q_1, q_2, q_3\}$

$\Sigma = \{0, 1\}$ $F = \{q_3\}$

Transition table

State	0	1
→ q_0		
q_1		ϕ
q_2	ϕ	
* q_3		

State transitions

q_0 loops on 0 and 1, with $1 \rightarrow q_1$

$q_1 \xrightarrow{0} q_2$

$q_2 \xrightarrow{1} q_3$

q_3 loops on 0 and 1

Sample Verification

101	Accepted
01010	Accepted
111	Rejected
100	Rejected.

Task 4:

$A = \{q_0\}$

Finding reachable subsets

$A = \{q_0\}$

$B = \{q_0, q_1\}$

$C = \{q_0, q_2\}$

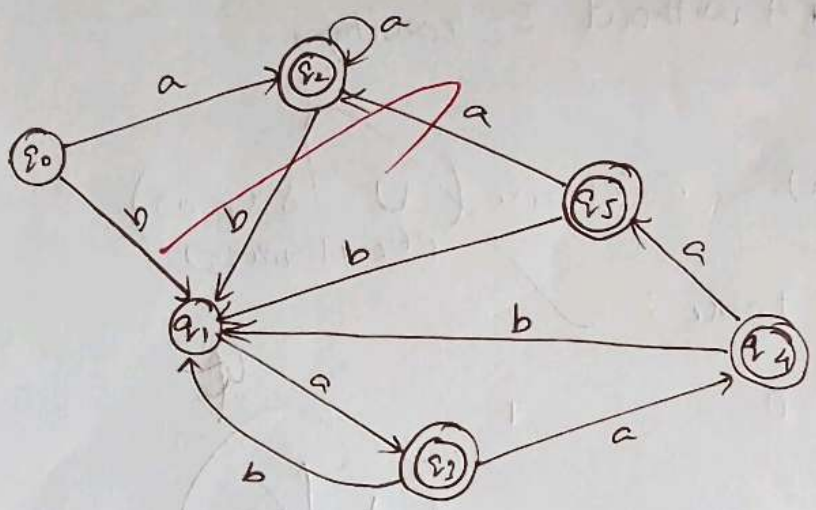
$D = \{q_0, q_1, q_3\}$

$E = \{q_0, q_2, q_4\}$

$F = \{q_0, q_3\}$

Subset construction table.

DFA state	0	1
→ A	A	B
B	C	B
C	A	D
* D	E	D
* E	F	D
* F	F	D



101 = Accepted

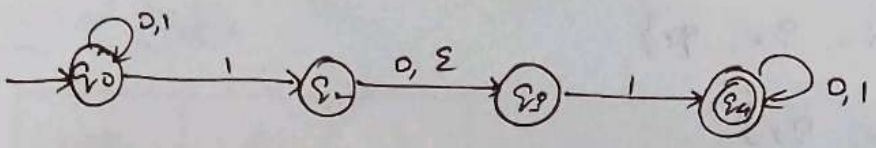
1101 = Accepted

100, 111 } Rejected.

Task: 5 ε - NFA construction

$Q = \{q_0, q_1, q_2, q_3, q_4\}$

State Transition Diagram



ε- NFA Transition table

State	0	1	ε
→q_0			
q_1	ϕ		ϕ
q_2	ϕ		ϕ
q_3		ϕ	ϕ
*q_4			ϕ

Equivalent NFA without ε- transitions

Using:

$$\delta'(q,a) = \epsilon\text{-closure}\left(\bigcup_{p \in \epsilon\text{-closure}(q)} \delta(p,a)\right)$$

New transition table:

State	0	1
→ q_0		
q_1	ϕ	
q_2	ϕ	
q_3		ϕ
* q_4		

Verification:

1110 } Accepted ; 101 } Rejected
0110 } Accepted ; 100 } Rejected